Chapter 4: Development of Graphical User Interface (GUI)

Chapter Importance (Exam Point of View)

- 4 Short Questions → 8 marks
- 2 MCQs → 2 marks
- Total → 10 marks from this chapter

Student Learning Outcomes

By the end of this chapter, students will be able to:

- Design interactive GUI-based programs using the Tkinter library.
- Connect Python applications to databases and perform CRUD operations.

Focus of the chapter:

- learn to develop user-friendly Graphical User Interfaces (GUIs) using Tkinter, Python's built-in GUI toolkit.
- Create windows, frames, labels, buttons, entry fields, and manage layouts using pack(), grid(), and place().
- Practise event handling and develop a simple login form.
- Learn database concepts, SQL, Python database connectivity, and basic CRUD operations using SQLite or MySQL.

4.1 Graphical User Interface (GUI) Development with Tkinter

GUI development

- Graphical User Interface (GUI) development allows programs to interact with users through visual elements like windows, buttons, labels, and text boxes (as shown in Figure 4.1).

Why GUI Development important in application development?

- GUIs make applications more user-friendly and easier to use compared to command-line interfaces.

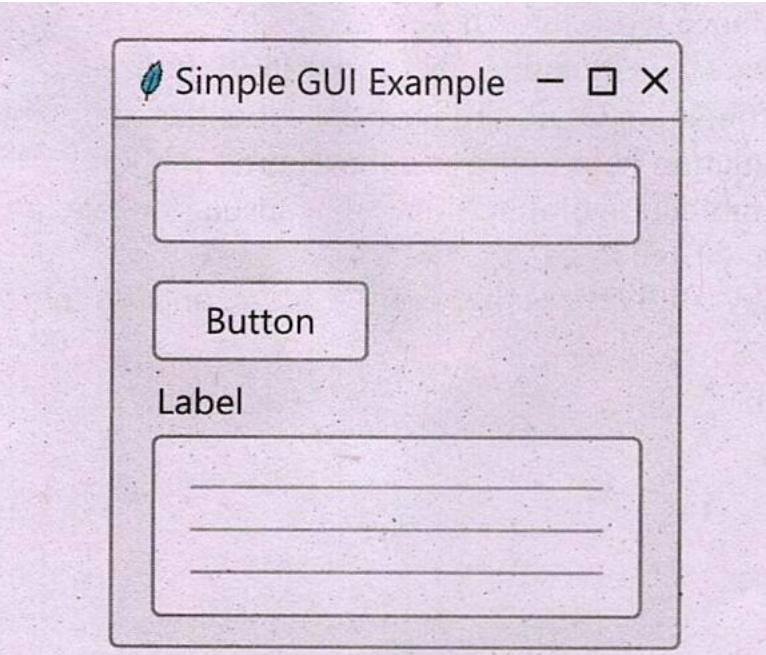


Figure 4.1: Graphical User Interface (GUI)

Tkinter - Python's built-in library

Python provides tools to create GUI applications efficiently. Tkinter is Python's built-in library for creating GUI programs. It is simple, widely used, and allows developers to design interactive applications quickly.

What is a GUI? Why is it important?

A **GUI (Graphical User Interface)** is a visual interface that allows users to interact with software using elements like buttons, menus, forms, and text boxes.

Importance Of GUIs:

- GUI makes programs easier to understand and operate (especially for non-technical users).
- They improve usability by providing visual feedback.
- It Reduces the need to remember complex commands.

GUI applications are widely used in desktop software, web applications, and mobile apps.

Overview of Tkinter

- Tkinter is Python’s standard built-in GUI toolkit.
- It comes pre-installed with Python.

- It provides a set of classes and functions to create windows, buttons, labels, entry fields, and other widgets.
- Tkinter is easy to learn, lightweight, and supports event-driven programming where actions are triggered by user events
- It is suitable for building small to medium-sized GUI applications quickly and efficiently.

Example:

Code:

```
import tkinter as tk

# Create main window
window = tk.Tk()
window.title("Simple GUI Example")
window.geometry("300x200")

# Function to handle button click (event)
def on_click():
    label.config(text="Button Clicked!")

# Create a label
label = tk.Label(window, text="Welcome to Tkinter")
label.pack(pady=10)

# Create an entry field
entry = tk.Entry(window)
entry.pack(pady=10)

# Create a button
button = tk.Button(window, text="Click Me", command=on_click)
button.pack(pady=10)

# Run the application
window.mainloop()
```

Output:



Basic Example Structure:

- Tk() → creates the main window
- Label → displays text
- Entry → takes user input
- Button → performs an action
- command=on_click → event handling (button click)
- mainloop() → runs the GUI application